20XT87 - DATA MINING LAB

PACKAGE ABSTRACT

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Introduction:

Accurate time-series forecasting is critical for modern infrastructure, ranging from traffic management to cloud resource allocation. While Graph Neural Networks (GNNs) have shown promise in modeling inter-node dependencies, they are inherently limited by their ability to only capture pairwise (one-to-one) relationships. Furthermore, most existing models rely on static graph structures that fail to account for the evolving and uncertain nature of real-world interactions.

Description:

This project implements a novel deep learning framework: **Probabilistic Hypergraph Recurrent Neural Networks (PHRNN)**. The core contribution of this work lies in replacing traditional graph edges with **hyperedges**, which allow for the modeling of "beyond-pairwise" group interactions (broadcast-style communication). Unlike deterministic models, the PHRNN architecture learns a **probabilistic hypergraph structure** from raw time-series data using the **Gumbel-Softmax reparameterization trick**. This allows the model to capture dynamic, fluctuating strengths in node-to-group connections while remaining end-to-end differentiable.

The implementation follows a four-stage architectural flow:

1. **Feature Extraction:** Utilizing 1D-Convolutional layers to capture local temporal patterns.
2. **Structural Initialization:** Leveraging K-Nearest Neighbors (KNN) to establish a prior-knowledge hypergraph.

3. **Spatial-Temporal Processing:** Deploying a custom PHRNN cell that performs two-stage aggregation (Node-to-Hyperedge and Hyperedge-to-Node) to update hidden states.
4. **Forecasting:** An Encoder-Decoder setup to generate multi-step ahead predictions.

Datasets Used:

The project is evaluated using the **Google Cloud Trace** and **Traffic Occupancy** datasets. Success is measured by comparing Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) against baseline models such as STGCN and MTGNN. The expected outcome is a robust forecasting system capable of autonomously mining complex group dependencies, thereby achieving state-of-the-art accuracy in highly dynamic environments.